

Problem

Solve for $v(t)$ in the integrodifferential equation

$$12 \frac{dv}{dt} + 36 \int_0^t v d\tau = 0$$

given that $v(0) = 2$.

Step-by-step solution

Step 1 of 2

The integrodifferential equation is,

$$12 \frac{dv}{dt} + 36 \int_0^t v dt = 0$$

Apply Laplace transform.

$$12L\left\{\frac{dv}{dt}\right\} + 36L\left\{\int_0^t v dt\right\} = 0$$

$$12[sV(s) - v(0)] + 36\left[\frac{V(s)}{s}\right] = 0$$

Substitute 2 for $v(0)$ in the above equation.

$$12[sV(s) - 2] + 36\left[\frac{V(s)}{s}\right] = 0$$

$$\left(12s + \frac{36}{s}\right)V(s) = 24$$

$$\left(\frac{12s^2 + 36}{s}\right)V(s) = 24$$

$$V(s) = \frac{24s}{12(s^2 + 3)}$$

$$= \frac{2s}{s^2 + 3}$$

Step 2 of 2

It is known that,

$$L^{-1}\left\{\frac{s}{s^2 + a^2}\right\} = \cos at$$

The simplified expression is,

$$V(s) = \frac{2s}{s^2 + 3}$$

$$= \frac{2s}{s^2 + (\sqrt{3})^2}$$

Apply inverse Laplace transform.

$$v(t) = 2 \cos(\sqrt{3}t)$$

Therefore, the voltage expression, $v(t)$ is $\boxed{2 \cos(\sqrt{3}t)}$.

